

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCOE504A - HUMAN COMPUTER INTERACTION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12**
- 1 Which of the following is the primary goal of Human Computer Interaction (HCI)?
 - a) To design user-friendly interfaces
 - b) To make computers faster
 - c) To reduce software size
 - d) To increase hardware requirements
 - 2 Navigation design primarily focuses on _____.
 - a) Determining software architecture
 - b) Helping users move through the interface easily
 - c) Increasing network speed
 - d) Improving security protocols
 - 3 “Visibility of system status” means the system should _____.
 - a) Hide internal operations
 - b) Run silently
 - c) Provide clear and timely feedback
 - d) Never show error messages
 - 4 Which of the following is a key element of a windowing system?
 - a) Windows, icons, menus and pointers
 - b) CPU scheduling
 - c) Database connectivity
 - d) Machine code translation
 - 5 Nielsen’s heuristics mainly focus on _____.
 - a) Hardware architecture
 - b) User interface usability
 - c) Data structures
 - d) Programming languages

- 6 Which is a major advantage of face-to-face communication?
 - a) Always recorded automatically
 - b) Requires no interpretation
 - c) Supports immediate feedback
 - d) Eliminates misunderstandings completely
- 7 Computer-Mediated Communication (CMC) involves communication through _____.
 - a) Physical gestures only
 - b) Electronic devices and networks
 - c) Face-to-face meetings
 - d) Printed documents
- 8 A mouse is an example of which type of device?
 - a) Text entry device
 - b) Storage device
 - c) Display device
 - d) Positioning, pointing device
- 9 Evaluation through user participation focuses on _____.
 - a) What experts believe about usability
 - b) Direct feedback from end users
 - c) Only error logging
 - d) System performance benchmarking
- 10 In HCI, “reasoning” refers to _____.
 - a) Remembering instructions
 - b) Drawing conclusions from available information
 - c) Performing random actions
 - d) Storing sensory data
- 11 A physical model represents _____.
 - a) The user's physical actions when interacting with devices
 - b) Hardware repairs
 - c) AI decision-making
 - d) Operating system schedules
- 12 Which of the following is an example of ubiquitous computing application?
 - a) Smart home automation
 - b) Fax machines
 - c) Typewriters
 - d) CRT monitors

Q2. Solve the following.

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|---|---|
| A) Write a short note on Human Memory | 6 |
| B) Explain in brief Reasoning and Problem Solving in HCI. | 6 |

Q3. Solve the following.

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|---|---|
| A) State & explain golden rule of design. | 6 |
| B) Explain Software Process in HCI | 6 |

Q4. Solve Any Two of the following.

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|---|---|
| A) State the Universal Design Principles in detail. | 6 |
| B) Write briefly about Nielsen’s heuristics for evaluation. | 6 |
| C) Explain different element of Windowing System. | 6 |

- Q5. Solve Any Two of the following.
- A) Explain Face to Face communication in detail. 6
 - B) Explain Keystroke-level model. 6
 - C) Write a note on GOMS. 6
- Q6. Solve Any Two of the following.
- A) Explain the concept of Augmented Reality (AR) and its applications in various domains 6
 - B) Compare Static and Dynamic web content. 6
 - C) Explain Computer Mediated Communication in detail. 6

***** End *****